

- 6 Since the rod is balanced at the centre, Sum of Clockwise moment = Sum of Anticlockwise moment. Hence, tension in both strings supporting P and Q are equal.

By drawing FBD of P, we can obtain: $T + V\rho_x g = m_P g$ --- (1)

By drawing FBD of Q, we can obtain: $T + V\rho_y g = m_Q g$ --- (2)

Taking (1) – (2),

$$\rightarrow V\rho_y g - V\rho_x g = m_Q g - m_P g$$

$$\rightarrow V\rho_y - V\rho_x = m_Q - m_P \text{ --- (3)}$$

Since $\rho_y > \rho_x$,

LHS of (3) is positive.

Therefore, $0 < m_Q - m_P \rightarrow m_Q < m_P$